

Conditional Independence Tests: Decision Tree

Use this notation with three variables:

X = candidate variable Z = conditioning variable Y = target

Step 1 - Test before conditioning

First ask whether X and Y are associated before conditioning on anything:

$$T_A: p(y \mid x) \stackrel{?}{=} p(y)$$

If $p(y \mid x) \neq p(y)$, then X and Y are dependent before conditioning.

If $p(y \mid x) = p(y)$, then X and Y are independent before conditioning.

Step 2A - If X and Y are already dependent

Now test whether conditioning on Z removes the dependence:

$$T_B: p(y \mid x, z) \stackrel{?}{=} p(y \mid z)$$

If $p(y \mid x, z) = p(y \mid z)$, then Z blocks the X - Y relationship.

Typical causal possibilities:

$X \rightarrow Z \rightarrow Y$ mediator

$X \leftarrow Z \rightarrow Y$ confounder

Probability alone usually cannot distinguish mediator from confounder. You need process order, time order, intervention logic, or domain knowledge.

If $p(y \mid x, z) \neq p(y \mid z)$, then the relationship survives adjustment. Possible explanations include direct effect, partial mediation, residual confounding, or hidden confounding.

Step 2B - If X and Y are initially independent

Use the same conditional test, but now the starting branch is different:

$$T_C: p(y \mid x, z) \stackrel{?}{=} p(y \mid z)$$

If $p(y \mid x, z) \neq p(y \mid z)$, then conditioning on Z has created dependence.

This is the collider pattern:

$X \rightarrow Z \leftarrow Y$

If $p(y \mid x, z) = p(y \mid z)$, then X and Y remain independent after conditioning on Z .

Compact notation

Define:

$D0(X, Y) = 1$ means X and Y are dependent before conditioning

$DZ(X, Y) = 1$ means X and Y are dependent after conditioning on Z

Before	After conditioning on Z	Interpretation
$D0 = 1$	$DZ = 0$	Z blocks the path: mediator or confounder candidate
$D0 = 1$	$DZ = 1$	Relationship survives: direct effect, partial mediation, residual or hidden confounding
$D0 = 0$	$DZ = 1$	Collider pattern: conditioning created dependence

D0 = 0	DZ = 0	No revealed relationship
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Key idea

The probability comparison $p(y | x, z) \neq p(y | z)$ is reused in different branches. What changes is the result before conditioning: whether X and Y were already dependent or initially independent.